PUSHFORWARDS OF ADJOINT BUNDLES WITH NEF TWISTS NEED NOT BE GV-SHEAVES

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Abstract. We answer a question of Meng negatively: there exist a smooth morphism with connected fibers from a smooth projective fourfold to an abelian surface and a nef line bundle on the fourfold such that the pushforward of the canonical bundle of the fourfold twisted by the nef line bundle is not a GV-sheaf.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $f : X \to A$ be a morphism from a smooth projective variety X to an abelian variety A. Popa and Schnell [PS14, Section 1] describe a theorem of Hacon [Hac04, Corollary 4.2(1)] in the following form: for any integer $i \geq 0$,

$$R^i f_* \mathcal{O}_X(K_X) \text{ is a GV-sheaf on } A$$

(see Definition 2.1). Popa and Schnell [PS14, Theorem 1.10] proved that for any positive integer m,

$$f_* \mathcal{O}_X(mK_X) \text{ is a GV-sheaf on } A.$$

Meng asked the following question.

Question 1.1 ([Men21a, Section 1])**.** Let X be a smooth projective variety, let $f : X \to A$ be a morphism to an abelian variety A, let m be a positive integer, and let L be a nef Cartier divisor on X. Is $f_* \mathcal{O}_X(mK_X + L)$ a GV-sheaf?

Besides the case $L = 0$ above, Question 1.1 has a positive answer in the following cases:

(1) When f is a contraction and $L = mM$ for a nef and f-big Cartier divisor M on X [PS14].
(2) When L is nef and f-abundant (see [Men21a, Section 2] for the definition) [Men21a].

The result of [Men21a] in (2) fails if L is only assumed to be f-nef and f-abundant [Men21a, Section 4]. To our knowledge, no answer to Question 1.1 for arbitrary nef L was known. The goal of this paper is to show that the answer to Question 1.1 is negative already when $m = 1$.

In Theorem 1.2, $\mathrm{Pic}^0(A)$ and $V^i(A, \mathcal{F})$ are as in Definition 2.1.

Theorem 1.2. *Let E be an elliptic curve, let $A := E \times E$, and let C be a smooth projective curve of genus 3. Then there exist a locally free sheaf $\mathcal{E}$ of rank 2 on $C \times A$ and a nef Cartier divisor L on $X := \mathbb{P}(\mathcal{E})$ with the following property. Let $f : X \to A$ be the composition of the projections $X \to C \times A \to A$, and let $\mathcal{F} := f_* \mathcal{O}_X(K_X + L)$. Then*

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(1) X is a smooth projective variety of dimension 4 and f is a smooth contraction,
(2) $\mathcal{F}$ is locally free of rank 5, and
(3) for any $\alpha \in \mathrm{Pic}^0(A)$,

$$h^1(A, \mathcal{F} \otimes \alpha) \geq 3 \quad \textit{and} \quad H^2(A, \mathcal{F} \otimes \alpha) = 0.$$

In particular, $V^1(A, \mathcal{F}) = \mathrm{Pic}^0(A)$, $V^i(A, \mathcal{F}) = \emptyset$ for any $i \geq 2$, and $\mathcal{F}$ is not a GV-sheaf.

Since $\mathcal{F}$ is not a GV-sheaf, [Men21a, Theorem 1.4] implies that L is not f-abundant.

Sketch of the proof. Fix the origin $0 \in E$, and let

$$H := \{0\} \times E + E \times \{0\} + \{(x, x) \mid x \in E\}.$$

Then $\mathcal{O}_A(H)$ is globally generated (Construction-Proposition 3.1), and for any $\alpha \in \mathrm{Pic}^0(A)$ and any integer $i > 0$,

$$h^0(A, \mathcal{O}_A(H) \otimes \alpha) = 3, \quad h^0(A, \mathcal{O}_A(2H) \otimes \alpha) = 12, \quad \text{and}$$
$$H^i(A, \mathcal{O}_A(H) \otimes \alpha) = H^i(A, \mathcal{O}_A(2H) \otimes \alpha) = 0.$$

Let $Z := C \times A$, and let $g : Z \to A$ and $h : Z \to C$ be the projections. Since $h^0(C, \omega_C) = h^0(A, \mathcal{O}_A(H)) = 3$, we choose a basis v_1, v_2, v_3 of $H^0(C, \omega_C)$ and a basis s_1, s_2, s_3 of $H^0(A, \mathcal{O}_A(H))$. Let e_1, e_2, e_3 be the basis of $H^1(C, \mathcal{O}_C)$ with $\mathrm{tr}(v_j \cup e_i) = \delta_{ij}$ for any $i, j \in \{1, 2, 3\}$, where $\mathrm{tr} : H^1(C, \omega_C) \to \mathbb{C}$ is the trace map, and let $\mathcal{E}$ be the extension

$$0 \to g^*\mathcal{O}_A(H) \to \mathcal{E} \to \mathcal{O}_Z \to 0$$

with class

$$\xi := \sum_{i=1}^{3} h^* e_i \cup g^* s_i \in H^1(Z, g^*\mathcal{O}_A(H))$$

(Construction 3.2). For $X = \mathbb{P}(\mathcal{E})$ and L with $\mathcal{O}_X(L) \cong \mathcal{O}_X(3)$,

$$\mathcal{F} \cong \mathcal{O}_A(H) \otimes g_*(h^*\omega_C \otimes \mathcal{E})$$

(see (3.2)). Applying $g_*(h^*\omega_C \otimes -)$ to the extension above, we obtain the exact sequence

$$0 \to H^0(C, \omega_C) \otimes \mathcal{O}_A(H) \to g_*(h^*\omega_C \otimes \mathcal{E}) \to H^0(C, \omega_C) \otimes \mathcal{O}_A \xrightarrow{\partial} \mathcal{O}_A(H),$$

whose connecting map ∂ satisfies $\partial(v_j \otimes 1) = s_j$ for any $j \in \{1, 2, 3\}$ (see (3.4) and (3.5)). Let $\mathcal{M}_H$ be the kernel of the evaluation map

$$H^0(A, \mathcal{O}_A(H)) \otimes \mathcal{O}_A \to \mathcal{O}_A(H).$$

Thus $\mathrm{Ker}\,\partial \cong \mathcal{M}_H$ and

$$0 \to H^0(C, \omega_C) \otimes \mathcal{O}_A(2H) \to \mathcal{F} \to \mathcal{M}_H \otimes \mathcal{O}_A(H) \to 0$$

(see (3.7)).

For any $\alpha \in \mathrm{Pic}^0(A)$, $H^2(A, \mathcal{F} \otimes \alpha) = 0$ and $H^1(A, \mathcal{F} \otimes \alpha)$ is the cokernel of the multiplication map

$$H^0(A, \mathcal{O}_A(H)) \otimes H^0(A, \mathcal{O}_A(H) \otimes \alpha) \to H^0(A, \mathcal{O}_A(2H) \otimes \alpha)$$

(see (3.8) and the vanishing in Construction-Proposition 3.1). Hence

$$h^1(A, \mathcal{F} \otimes \alpha) \geq h^0(A, \mathcal{O}_A(2H) \otimes \alpha) - h^0(A, \mathcal{O}_A(H)) \cdot h^0(A, \mathcal{O}_A(H) \otimes \alpha) = 12 - 3 \cdot 3 = 3.$$

Structure of the paper. In Section 2 we recall the definition of GV-sheaves and its equivalence with the definition used in [Men21a]. In Section 3 we construct $\mathcal{E}$ and L and prove the main theorem.

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2. GV-sheaves

Definition 2.1 ([PP11, Proposition/Definition 2.1])**.** Let A be an abelian variety, let $\mathrm{Pic}^0(A)$ be the dual abelian variety of A [Mum74, §8], whose points we identify with the corresponding line bundles on A, and let $\mathcal{F}$ be a coherent sheaf on A. For any integer $i \geq 0$, let

$$V^i(A, \mathcal{F}) := \{\alpha \in \mathrm{Pic}^0(A) \mid H^i(A, \mathcal{F} \otimes \alpha) \neq 0\}.$$

We say that $\mathcal{F}$ is a *GV-sheaf* if $\operatorname{codim}_{\mathrm{Pic}^0(A)} V^i(A, \mathcal{F}) \geq i$ for any $i > 0$, with the convention $\operatorname{codim} \emptyset = +\infty$.

By [Men21a, Section 4], the GV-sheaves in Question 1.1 are those of [Men21b, Definition 2.4(i)]: a coherent sheaf $\mathcal{F}$ on an abelian variety A is a GV-sheaf if

$$\operatorname{codim}_{\mathrm{Pic}^0(A)} \operatorname{Supp} R^i\widehat{\mathcal{S}}\mathcal{F} \geq i \quad \text{for any } i > 0, \tag{2.1}$$

where $R^i\widehat{\mathcal{S}}\mathcal{F}$ is the i-th cohomology sheaf of the Fourier–Mukai transform of $\mathcal{F}$ with respect to the normalized Poincaré bundle on $A \times \mathrm{Pic}^0(A)$. By [PP11, Proposition/Definition 2.1], the inequalities (2.1) hold if and only if $\mathcal{F}$ is a GV-sheaf in the sense of Definition 2.1.

3. Proof of Theorem 1.2

The sheaf $\mathcal{E}$ that we construct for Theorem 1.2 is an extension

$$0 \to g^*\mathcal{O}_A(H) \to \mathcal{E} \to \mathcal{O}_{C\times A} \to 0$$

for the projection $g : C \times A \to A$ and a globally generated ample line bundle $\mathcal{O}_A(H)$ on A with $h^0(A, \mathcal{O}_A(H)) = 3 = h^0(C, \omega_C)$, so we first construct $\mathcal{O}_A(H)$.

Construction-Proposition 3.1. Let E be an elliptic curve with origin 0, let $A := E \times E$, and consider the Cartier divisor

$$H := \{0\} \times E + E \times \{0\} + \{(x, x) \mid x \in E\}$$

on A. Then

(1) $\mathcal{O}_A(H)$ is globally generated, and
(2) for any $\alpha \in \mathrm{Pic}^0(A)$ and any integer $i > 0$,

$$h^0(A, \mathcal{O}_A(H) \otimes \alpha) = 3, \quad h^0(A, \mathcal{O}_A(2H) \otimes \alpha) = 12, \quad \text{and}$$
$$H^i(A, \mathcal{O}_A(H) \otimes \alpha) = H^i(A, \mathcal{O}_A(2H) \otimes \alpha) = 0.$$

Proof. Let

$$D_1 := \{0\} \times E, \quad D_2 := E \times \{0\}, \quad \text{and} \quad D_3 := \{(x, x) \mid x \in E\},$$

so that $H = D_1 + D_2 + D_3$. Let $\mathrm{pr}_1, \mathrm{pr}_2 : A \to E$ be the projections to the first and the second factor, and let $\mu : A \to E$ be the morphism given by $\mu(x, y) = x - y$. Since D_1, D_2, and D_3 are fibers of pr_1, pr_2, and μ meeting pairwise transversally at $(0, 0)$ only, $(H^2) = 6$. As $D_1 + D_2$ is ample and D_3 is nef, H is ample.

We prove (2). Let $\alpha \in \mathrm{Pic}^0(A)$. Since α is algebraically equivalent to 0 and H is ample, $\mathcal{O}_A(H) \otimes \alpha$ and $\mathcal{O}_A(2H) \otimes \alpha$ are ample. Since $\omega_A \cong \mathcal{O}_A$, by Kodaira vanishing,

$$H^i(A, \mathcal{O}_A(H) \otimes \alpha) = H^i(A, \mathcal{O}_A(2H) \otimes \alpha) = 0 \quad \text{for any integer } i > 0.$$

Since α is numerically trivial, by the Riemann–Roch theorem,

$$h^0(A, \mathcal{O}_A(H) \otimes \alpha) = \chi(A, \mathcal{O}_A(H) \otimes \alpha) = \frac{(H^2)}{2} = 3, \quad h^0(A, \mathcal{O}_A(2H) \otimes \alpha) = \frac{((2H)^2)}{2} = 12.$$

We prove (1). For any $t \in E$, let

$$H_t := \mathrm{pr}_1^{-1}(-t) + \mathrm{pr}_2^{-1}(t) + \mu^{-1}(t).$$

Then $H_0 = H$. For any $t \in E$, let $\mathcal{P}_t := \mathcal{O}_E(t-0) \in \mathrm{Pic}^0(E)$, so that $\mathcal{O}_E(t) \cong \mathcal{O}_E(0) \otimes \mathcal{P}_t$. Since $t \to \mathcal{O}_E(t-0)$ is a group isomorphism $E \to \mathrm{Pic}^0(E)$, $\mathcal{P}_{-t} \cong \mathcal{P}_t^{-1}$. By [Mum74, §8, (ii) and (iii)],

$$\mu^*\mathcal{P}_t \cong \mathrm{pr}_1^*\,\mathcal{P}_t \otimes \mathrm{pr}_2^*\,\mathcal{P}_t^{-1}.$$

Thus

$$\begin{aligned}\mathcal{O}_A(H_t) &\cong \mathcal{O}_A(H) \otimes \mathrm{pr}_1^*\,\mathcal{P}_{-t} \otimes \mathrm{pr}_2^*\,\mathcal{P}_t \otimes \mu^*\mathcal{P}_t \\ &\cong \mathcal{O}_A(H) \otimes \mathrm{pr}_1^*(\mathcal{P}_t^{-1} \otimes \mathcal{P}_t) \otimes \mathrm{pr}_2^*(\mathcal{P}_t \otimes \mathcal{P}_t^{-1}) \cong \mathcal{O}_A(H),\end{aligned}$$

that is, $H_t \in |H|$. Let $(x, y) \in A$. Since E is infinite, we may pick $t \in E\backslash\{-x, y, x-y\}$. Then $x \neq -t$, $y \neq t$, and $x - y \neq t$, so $(x, y) \notin H_t$. Thus the section of $\mathcal{O}_A(H)$ defining H_t does not vanish at (x, y), and $\mathcal{O}_A(H)$ is globally generated. □

Construction 3.2. Notation as in Construction-Proposition 3.1. Let C be a smooth projective curve of genus 3, let $Z := C \times A$, and let $g : Z \to A$ and $h : Z \to C$ be the projections. Let v_1, v_2, v_3 be a basis of $H^0(C, \omega_C)$, let e_1, e_2, e_3 be the basis of $H^1(C, \mathcal{O}_C)$ with $\mathrm{tr}(v_j \cup e_i) = \delta_{ij}$ for any $i, j \in \{1, 2, 3\}$, where $\mathrm{tr} : H^1(C, \omega_C) \to \mathbb{C}$ is the trace map, and let s_1, s_2, s_3 be a basis of $H^0(A, \mathcal{O}_A(H))$. We let

(1) $\mathcal{E}$ be the locally free sheaf of rank 2 on Z given by the extension

$$0 \to g^*\mathcal{O}_A(H) \to \mathcal{E} \to \mathcal{O}_Z \to 0 \tag{3.1}$$

whose class is

$$\xi := \sum_{i=1}^{3} h^*e_i \cup g^*s_i \in H^1(Z, g^*\mathcal{O}_A(H)) = \mathrm{Ext}^1(\mathcal{O}_Z, g^*\mathcal{O}_A(H)),$$

(2) $X := \mathbb{P}(\mathcal{E})$ be the projective bundle of 1-dimensional quotients of $\mathcal{E}$, with projection $\pi : X \to Z$, so that $\pi_*\mathcal{O}_X(1) = \mathcal{E}$,
(3) $f := g \circ \pi : X \to A$,
(4) L be a Cartier divisor on X with $\mathcal{O}_X(L) \cong \mathcal{O}_X(3)$, and
(5) $\mathcal{F} := f_*\mathcal{O}_X(K_X + L)$.

Proof of Theorem 1.2. Fix a point $0 \in E$, and let H, Z, g, h, $\mathcal{E}$, X, π, f, L, and $\mathcal{F}$ be as in Construction 3.2. Then $f = g \circ \pi$ is the composition of the projections $X \to C \times A \to A$. We show that L is nef and that (1)–(3) hold.

Step 1. In this step we prove (1) and that L is nef. Since π is a $\mathbb{P}^1$-bundle over the smooth projective threefold Z and g is a smooth contraction, (1) holds.

By Construction-Proposition 3.1(1), $g^*\mathcal{O}_A(H)$ is globally generated, hence nef. Since $\mathcal{O}_Z$ is nef, by [Laz04, Theorem 6.2.12(ii)], the extension $\mathcal{E}$ is a nef vector bundle, that is, $\mathcal{O}_X(1)$ is nef [Laz04, Definition 6.1.1]. Since $\mathcal{O}_X(L) \cong \mathcal{O}_X(1)^{\otimes 3}$, L is nef.

Step 2. In this step we show that $\mathcal{F} \cong \mathcal{O}_A(H) \otimes g_*(h^*\omega_C \otimes \mathcal{E})$. Since $\omega_{X/Z} \cong \mathcal{O}_X(-2) \otimes \pi^* \det \mathcal{E}$, $\det \mathcal{E} \cong g^*\mathcal{O}_A(H)$, and $\omega_Z \cong h^*\omega_C$ (as $\omega_A \cong \mathcal{O}_A$), we have

$$\mathcal{O}_X(K_X + L) \cong \mathcal{O}_X(1) \otimes \pi^*(h^*\omega_C \otimes g^*\mathcal{O}_A(H)).$$

Since $\pi_*\mathcal{O}_X(1) = \mathcal{E}$, by the projection formula for π and then for g,

$$\mathcal{F} \cong \mathcal{O}_A(H) \otimes \mathcal{G}, \quad \text{where } \mathcal{G} := g_*(h^*\omega_C \otimes \mathcal{E}). \tag{3.2}$$

Step 3. In this step we compute the connecting map ∂ in (3.4) below, and we prove (2). Tensoring (3.1) with $h^*\omega_C$, we obtain the exact sequence

$$0 \to h^*\omega_C \otimes g^*\mathcal{O}_A(H) \to h^*\omega_C \otimes \mathcal{E} \to h^*\omega_C \to 0. \tag{3.3}$$

By flat base change and the projection formula,

$$g_*(h^*\omega_C) \cong H^0(C, \omega_C) \otimes \mathcal{O}_A, \quad g_*(h^*\omega_C \otimes g^*\mathcal{O}_A(H)) \cong H^0(C, \omega_C) \otimes \mathcal{O}_A(H), \quad \text{and}$$
$$R^1g_*(h^*\omega_C \otimes g^*\mathcal{O}_A(H)) \cong H^1(C, \omega_C) \otimes \mathcal{O}_A(H) \cong \mathcal{O}_A(H),$$

where the last isomorphism is $\mathrm{tr}\otimes \mathrm{id}_{\mathcal{O}_A(H)}$. Applying g_* to (3.3), we obtain the exact sequence

$$0 \to H^0(C,\omega_C)\otimes \mathcal{O}_A(H) \to \mathcal{G} \to H^0(C,\omega_C)\otimes \mathcal{O}_A \xrightarrow{\partial} \mathcal{O}_A(H), \tag{3.4}$$

where ∂ is the connecting map. We show that

$$\partial(v_j\otimes 1) = s_j \quad \text{for any } j\in\{1,2,3\}. \tag{3.5}$$

Since $H^0(C,\omega_C)\otimes\mathcal{O}_A$ is free with basis $v_1\otimes 1, v_2\otimes 1, v_3\otimes 1$, where $v_j\otimes 1$ corresponds to $h^*v_j\in H^0(Z,h^*\omega_C)$, ∂ is determined by $H^0(A,\partial)$, and

$$H^0(A,\partial) = (\mathrm{tr}\otimes \mathrm{id}_{H^0(A,\mathcal{O}_A(H))})\circ\beta\circ\partial_Z,$$

where

$$\partial_Z : H^0(Z,h^*\omega_C) \to H^1(Z,h^*\omega_C\otimes g^*\mathcal{O}_A(H))$$

is the connecting map of (3.3) and

$$\beta : H^1(Z,h^*\omega_C\otimes g^*\mathcal{O}_A(H)) \to H^1(C,\omega_C)\otimes H^0(A,\mathcal{O}_A(H))$$

is the natural map. For any $v\in H^0(C,\omega_C)$, multiplication by h^*v maps (3.1) to (3.3), and ξ is the image of 1 under the connecting map $H^0(Z,\mathcal{O}_Z)\to H^1(Z,g^*\mathcal{O}_A(H))$ of (3.1). Thus, by the naturality of connecting maps,

$$\partial_Z(h^*v_j) = h^*v_j\cup\xi = \sum_{i=1}^{3} h^*(v_j\cup e_i)\cup g^*s_i.$$

Let

$$\beta_0 : H^1(Z,h^*\omega_C) \to H^1(C,\omega_C)\otimes H^0(A,\mathcal{O}_A)$$

be the natural map, so that $\beta_0(h^*c) = c\otimes 1$ for any $c\in H^1(C,\omega_C)$. Under the base change isomorphisms, multiplication by g^*s_i from $h^*\omega_C$ to $h^*\omega_C\otimes g^*\mathcal{O}_A(H)$ induces multiplication by s_i from $H^1(C,\omega_C)\otimes\mathcal{O}_A$ to $H^1(C,\omega_C)\otimes\mathcal{O}_A(H)$ (projection formula). The maps β_0 and β are compatible with the maps induced by multiplication by g^*s_i, so $\beta(h^*c\cup g^*s_i) = c\otimes s_i$ for any $c\in H^1(C,\omega_C)$, and

$$\beta(\partial_Z(h^*v_j)) = \sum_{i=1}^{3}(v_j\cup e_i)\otimes s_i.$$

Hence

$$H^0(A,\partial)(v_j\otimes 1) = (\mathrm{tr}\otimes \mathrm{id}_{H^0(A,\mathcal{O}_A(H))})(\beta(\partial_Z(h^*v_j))) = \sum_{i=1}^{3}\mathrm{tr}(v_j\cup e_i)s_i = s_j.$$

This proves (3.5). Thus ∂ is the composite of the isomorphism

$$H^0(C,\omega_C)\otimes\mathcal{O}_A \to H^0(A,\mathcal{O}_A(H))\otimes\mathcal{O}_A, \quad v_j\otimes 1\to s_j\otimes 1,$$

and the evaluation map

$$H^0(A,\mathcal{O}_A(H))\otimes\mathcal{O}_A\to\mathcal{O}_A(H).$$

Let $\mathcal{M}_H$ be the locally free sheaf of rank 2 in the exact sequence

$$0\to\mathcal{M}_H\to H^0(A,\mathcal{O}_A(H))\otimes\mathcal{O}_A\to\mathcal{O}_A(H)\to 0, \tag{3.6}$$

where the right map is the evaluation map, surjective by Construction-Proposition 3.1(1). By (3.4) and the isomorphism $v_j\otimes 1\to s_j\otimes 1$, we have the exact sequence

$$0\to H^0(C,\omega_C)\otimes\mathcal{O}_A(H)\to\mathcal{G}\to\mathcal{M}_H\to 0.$$

Thus $\mathcal{G}$ is locally free of rank $3+2=5$, and so is $\mathcal{F}$ by (3.2). This proves (2). Tensoring with $\mathcal{O}_A(H)$, by (3.2) we obtain the exact sequence

$$0\to H^0(C,\omega_C)\otimes\mathcal{O}_A(2H)\to\mathcal{F}\to\mathcal{M}_H\otimes\mathcal{O}_A(H)\to 0. \tag{3.7}$$

Step 4. In this step we prove (3). Let $\alpha \in \mathrm{Pic}^0(A)$. Tensoring (3.6) with $\mathcal{O}_A(H) \otimes \alpha$, we obtain the exact sequence

$$0 \to \mathcal{M}_H \otimes \mathcal{O}_A(H) \otimes \alpha \to H^0(A, \mathcal{O}_A(H)) \otimes \mathcal{O}_A(H) \otimes \alpha \to \mathcal{O}_A(2H) \otimes \alpha \to 0. \tag{3.8}$$

By Construction-Proposition 3.1(2), the middle and right terms of (3.8) and $H^0(C, \omega_C) \otimes \mathcal{O}_A(2H) \otimes \alpha$ have no cohomology in positive degrees. Thus $H^1(A, \mathcal{M}_H \otimes \mathcal{O}_A(H) \otimes \alpha)$ is the cokernel of the multiplication map

$$H^0(A, \mathcal{O}_A(H)) \otimes H^0(A, \mathcal{O}_A(H) \otimes \alpha) \to H^0(A, \mathcal{O}_A(2H) \otimes \alpha),$$

and $H^2(A, \mathcal{M}_H \otimes \mathcal{O}_A(H) \otimes \alpha) = 0$. Thus

$$h^1(A, \mathcal{M}_H \otimes \mathcal{O}_A(H) \otimes \alpha) \geq 12 - 3 \cdot 3 = 3.$$

Thus, by (3.7) tensored with α,

$$H^1(A, \mathcal{F} \otimes \alpha) \cong H^1(A, \mathcal{M}_H \otimes \mathcal{O}_A(H) \otimes \alpha),$$
$$H^2(A, \mathcal{F} \otimes \alpha) \cong H^2(A, \mathcal{M}_H \otimes \mathcal{O}_A(H) \otimes \alpha) = 0.$$

Thus $h^1(A, \mathcal{F} \otimes \alpha) \geq 3$ and $H^2(A, \mathcal{F} \otimes \alpha) = 0$. This proves (3).

By (3), $V^1(A, \mathcal{F}) = \mathrm{Pic}^0(A)$ and $V^2(A, \mathcal{F}) = \emptyset$. In particular,

$$\mathrm{codim}_{\mathrm{Pic}^0(A)} V^1(A, \mathcal{F}) = 0 < 1.$$

Since $\dim A = 2$, $H^i(A, \mathcal{F} \otimes \alpha) = 0$ for any $\alpha \in \mathrm{Pic}^0(A)$ and any $i \geq 3$, so $V^i(A, \mathcal{F}) = \emptyset$ for any $i \geq 3$. By Definition 2.1, $\mathcal{F}$ is not a GV-sheaf. The theorem follows. □

References

[Hac04] C. D. Hacon, *A derived category approach to generic vanishing*, J. Reine Angew. Math. **575** (2004), 173–187.

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. II*, Ergeb. Math. Grenzgeb. (3) **49**, Springer-Verlag, Berlin (2004).

[Men21a] F. Meng, *On vanishing and torsion-freeness results for adjoint pairs*, arXiv:2106.11883.

[Men21b] F. Meng, *Pushforwards of klt pairs under morphisms to abelian varieties*, Math. Ann. **380** (2021), 1655–1685.

[Mum74] D. Mumford, *Abelian varieties*, second edition, Tata Institute of Fundamental Research Studies in Mathematics, No. 5, Oxford University Press, London (1974).

[PP11] G. Pareschi and M. Popa, *Regularity on abelian varieties III: relationship with generic vanishing and applications*, Grassmannians, moduli spaces and vector bundles, Clay Math. Proc. **14**, Amer. Math. Soc., Providence, RI (2011), 141–167.

[PS14] M. Popa and C. Schnell, *On direct images of pluricanonical bundles*, Algebra Number Theory **8** (2014), no. 9, 2273–2295.

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